2. Sampling Theory:

Random Sampling:

A large collection of individuals or numerical data can be understand as a population

A finite subset of the population is called sample. The number of individuals in a sample is called sample size (n)

If the sample size $n \leq 30$ is called small sample otherwise it is a large sample.

The process of selecting a sample from the population is called sampling. The selection of an individual or item from the population in such a way that each has the same change of being selected is called random sampling

Sampling where a member of the population may be selected more than once is called as sampling with replacement otherwise without replacement

Simple Sampling is a special case of random sampling in which trials are independent and the probability of success is a constant.

Sampling Distribution

We have a different sample size 'n' drawn from the population. For each every sample ~~s~~ of size n. we compute quantities like mean, standard deviation (S D) etc. we group these characteristics according to their frequencies. This frequency distributions gives sampling distribution.

NOTE: The sampling distribution of large samples is assumed to be a normal distribution.

Sampling Distribution of the Mean:

Case (i): Random Sampling with replacement.

The items are drawn one by one and are put back to the population before the next ~~draw~~ draw. If N is the size of the finite population & 'n' is the size of the sample then we have ($\mu_{\bar{x}}$) of the frequency distribution of the sample mean will be equal to the population mean(μ)

i.e, $\boxed{\mu_{\bar{x}} = \mu}$

The variance $\sigma^2_{\bar{x}}$ of the frequency distribution of the sample mean will be equal to σ^2/n where σ^2 is the variance of the population

i.e, $\boxed{\sigma^2_{\bar{x}} = \sigma^2/n}$ $\left(\sigma^2_{\bar{x}} = \sigma^2/n\right)$

Case 2: Random Sampling without replacement.

The mean $\mu_{\bar{x}}$ of the frequency distribution of the sample mean will be equal to the population mean (μ) i.e. $\boxed{\mu_{\bar{x}} = \mu}$

The variance $\sigma^2_{\bar{x}}$ of the frequency distribution of the sample mean is equal to $\left(\frac{N-n}{N-1}\right)\frac{\sigma^2}{n}$ where σ^2 is the variance of the population

i.e. $\sigma^2_{\bar{x}} = \left[\frac{N-n}{N-1}\right]\frac{\sigma^2}{n} = c \cdot \frac{\sigma^2}{n}$ where $c = \frac{N-n}{N-1}$

Testing of Hypothesis:

In order to make a decision regarding the population using a sample of the population we make some assumptions referred to as hypothesis which may or may not be true.

The hypothesis formulated for the purpose of its rejection under the assumption that it is true is called the Null Hypothesis denoted by H_0

The complementary to the null hypothesis is called Alternative Hypothesis denoted by H_1

Errors:

In a test process there can be four possible situation of which two of the situations leads to the two types of errors.

	Accepting the hypothesis	Rejecting the hypothesis
Hypothesis True	Correct decision	Wrong decision Type I error
Hypothesis False	Wrong decision Type II error	Correct decision

Significance level :

The probability level, below which leads to the rejection of the hypothesis is known as the significance level.

This probability is conventionally fixed at 0.05 or 0.01 i.e, 5% or 1%. These are called significance level.

Test of Significance and Confidence intervals

Let us suppose that we have a normal population with mean 'μ' and S.D. 'σ'. If $\overline{X}$ is the sample mean of a random sample of size 'n'. the quantity Z defined by

$$\boxed{Z = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}}}$$ is called Standard Normal Variate (SNV)

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I: 5% of significance level is equal to 95% of confidence limit.

95% confidence that z lies b/w ± 1.96

$$-1.96 \le \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \le 1.96$$

OR $\bar{x} - 1.96\left(\frac{\sigma}{\sqrt{n}}\right) \le \mu \le 1.96 \frac{\sigma}{\sqrt{n}} + \bar{x}$

II; 1% of significance level is equal to 99% of Confidence limit.

99% confidence that z lies b/w ± 2.58

$$\bar{x} - 2.58\left(\frac{\sigma}{\sqrt{n}}\right) \le \mu \le 2.58 \frac{\sigma}{\sqrt{n}} + \bar{x}.$$

Test of Significance of Proportions.

(i) Mean of Proportions $= \frac{np}{n} = p$.

(ii) S.D. of the proportions $= \frac{\sqrt{npq}}{n} = \sqrt{\frac{pq}{n}}$

Hence $z = \frac{x-\mu}{\sigma} = \frac{x-np}{\sqrt{npq}}$

if $|z| > 2.58$ we conclude that the difference is highly significant & Reject the hypothesis.

Problems:

① A population consist of 3 numbers $\{1, 2, 3\}$ form the sampling distribution of the sample means in the case of

i) Random sample of size 2 with replacement

(ii) Random sample of size 2 without replacement.

$\Rightarrow$ $N = 3$ (finite population)

population mean $\mu = \frac{1+2+3}{3} = 2$.

population variance $\sigma^2 = \frac{1}{N}\sum(x-\mu)^2$

$$= \frac{1}{3}\left\{(1-2)^2 + (2-2)^2 + (3-2)^2\right\} = \frac{2}{3}$$

case (i): Random sample size 2 (i.e, $n=2$) with replacement

(1, 1) (1, 2) (1, 3); (2, 1) (2, 2) (2, 3); (3, 1) (3, 2) (3, 3)

i.e, $N^n = 3^2 = 9$.

The mean of these respectively

1, 1.5, 2, 1.5, 2, 2.5, 2, 2.5, 3.

x	1	1.5	2	2.5	3
f	1	2	3	2	1

$$\mu_{\bar{x}} = \frac{\sum fx}{\sum f} = \frac{1+3+6+5+3}{9} = 2.$$

$$\sigma_{\bar{x}}^2 = \frac{\sum f\cdot(x-\mu_{\bar{x}})^2}{\sum f} = \frac{1}{3}$$

$$\mu_{\bar{x}} = 2 = \mu \; ; \; \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} = \frac{2/3}{2} = \frac{1}{3}$$

Case 2: Random sample size 2 (i.e, $n=2$) without replacement

$$^{N}C_{n} = {}^{3}C_{2} = 3 \; ; \; (1,2)\ (2,3)\ (3,1)$$

means are 1.5, 2.5, 2.

x	1.5	2	2.5
f	1	1	1

$$\text{mean} = \mu_{\bar{x}} = \frac{1.5+2+2.5}{3} = 2$$

$$\text{Variance} = \sigma^2_{\bar{x}} = \frac{1}{3}\left\{(1.5-2)^2+(2.5-2)^2+(2-2)^2\right\} = \frac{1}{6}$$

$$\mu_{\bar{x}} = \mu = 2$$

$$\left[\frac{N-n}{N-1}\right]\frac{\sigma^2}{n} = \frac{(3-2)}{3-1}\cdot\frac{2/3}{2} = \underline{\frac{1}{6}} = \sigma^2_{\bar{x}}$$

(2) The weights of 1500 ball bearings are normally distributed with a mean of 635 gms ~~of~~ and S.D. of 1.36 gms. If 300 random sample of size 36 are drawn from this population. Determine the expected mean and S.D. of the sampling distribution of means if sampling is done:
(i) With replacement (ii) without replacement.

$\Rightarrow$ $N = 1500$, $\mu = 635$, $\sigma = 1.36$, $n = 36$

(i) with replacement: $\mu_{\bar{x}} = \mu = 635$.

$$\sigma^2_{\bar{x}} = \sigma^2/n \;\;\&\;\; \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = 1.36/\sqrt{36} = \underline{0.227}$$

(ii) without replacement: $\mu_{\bar{x}} = \mu = 635$

$$\sigma^2_{\bar{x}} = \frac{N-n}{N-1}\cdot\frac{\sigma^2}{n} = \frac{1500-36}{1500-1}\cdot\frac{(1.36)^2}{36} = 0.05.$$

$$\sigma_{\bar{x}} = \sqrt{0.05} = \underline{0.224}$$

(3) The daily wages of 3000 workers in a factory are normally distributed with mean equal to Rs. 68 and S.D. equal to Rs. 3. If 80 samples consisting of 25 workers each are obtained. what would be the mean and S.D. of the sampling distribution of means if sampling were done (a) with replacement (b) without replacement In how many samples will the mean is likely to be
(i) b/w Rs. 66.8 & Rs. 68.3
(ii) less than Rs. 66.4 ?

$\Rightarrow N = 3000, \; n = 25, \; \mu = 68, \; \sigma = 3.$

(a) With replacement : $\mu_{\bar{x}} = \mu = 68$.

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \Rightarrow \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{3}{\sqrt{25}} = \underline{0.6}.$$

(b) Without replacement : $\mu_{\bar{x}} = \mu = 68$

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}\left(\frac{N-n}{N-1}\right) = \frac{9}{25}\left(\frac{3000-25}{3000-1}\right) = \frac{9}{25}*\overset{(0.998)}{(\simeq 1)} = \frac{9}{25}$$

$$\sigma_{\bar{x}} = \sqrt{9/25} = \underline{0.6}$$

* Mean and S.D's are almost same in both the cases.

Hence $\mu_{\bar{x}} = 68$ & $\sigma_{\bar{x}} = 0.6$.

$$\therefore \; Z = \frac{\bar{X} - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{\bar{X} - 68}{0.6}$$

$$\Rightarrow \bar{X} = 0.6Z + 68$$

(i) $P(66.8 \le \bar{X} \le 68.3) = P(66.8 \le 0.6Z + 68 \le 68.3)$

$$= P(-2 < Z < 0.5) = A(0.5) + A(2)$$

$$= 0.4772 + 0.1915 = \underline{0.6687}$$

The expected number of samples of having mean b/w 66.8 to 68.3 is

$$0.6687 \times 80 = \underline{53.49} \simeq \underline{\underline{53}}$$

ii) $P(\bar{X} < 66.4) = P(0.6Z + 68 < 66.4)$

$= P(Z < -2.67)$

$= 0.5 - A(2.67) = 0.5 - 0.4962$

$= \underline{0.0038}$.

$\therefore$ The expected number of samples of 80 having mean less than 66.4 is $0.0038 \times 80 = \underline{0.304} \approx \underline{0}$

④ Certain tube manufactured by a company have mean life time of 800 hrs & S.D. of 60 hrs. Find the probability that a random sample of 16 tube taken from the group will have a mean life time (i) b/w 790 hrs. to 810 hrs. (ii) less than 785 hrs. (iii) more than 820 hrs (iv) b/w 770 hrs. & 830 hrs.

$\Rightarrow \mu = 800, \ \sigma = 60, \ n = 16$

$\mu_{\bar{x}} = \mu = 800, \ \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{60}{\sqrt{16}} = 15$

$\therefore Z = \frac{\bar{X} - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{\bar{X} - 800}{15}$

$\therefore \bar{X} = 15Z + 800$.

(i) $P(790 \le \bar{X} \le 810) = P(790 \le 15Z + 800 \le 810)$

$= P(-0.67 \le Z \le 0.67)$

$= 2A(0.67)$

$= 2(0.2486)$

$= \underline{0.4972}$

(ii) $P(\bar{x} < 785) = P(15Z + 800 < 785)$

$= P(Z \leq -1)$

$= 0.5 - A(1) = 0.5 - 0.3413$

$= \underline{0.1587}$

(iii) $P(\bar{X} > 820) = P(15Z + 800 > 820)$

$= P(Z > 1.33) = 0.5 - A(1.33)$

$= 0.5 - 0.4082 = \underline{0.0918}$

(iv) $P(770 \leq \bar{X} \leq 830) = P(770 \leq 15Z + 800 \leq 830)$

$= P(-2 < Z < 2) = 2A(2)$

$= 2(0.4772) = \underline{0.9544}$

⑤ Find the probability that in 100 tosses of a fair coin b/w 45% and 55% of the outcomes are heads.

$\Rightarrow \mu_p = P = \frac{1}{2}$; $\sigma = \sqrt{\frac{pq}{n}}$. $n = 100$.

$q = 1 - p = \frac{1}{2}$; $\sigma = \sqrt{\frac{\frac{1}{2} \cdot \frac{1}{2}}{100}} = \frac{1}{20} = \underline{0.05}$

$$Z = \frac{x_p - \mu_p}{\sigma_p} = \frac{x_p - 0.5}{0.05}$$

$\therefore x_p = 0.05Z + 0.5$.

$\therefore P(0.45 < x_p < 0.55) = P(0.45 < 0.05Z + 0.5 < 0.55)$

$= P(-1 < Z < 1) = 2A(1)$

$= 2(0.3413)$

$= \underline{0.6826}$